



Top 14 CAT Simple Interest and Compound Interest Questions

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Questions

Instructions

For the following questions answer them individually

Question 1

Anil borrows Rs 2 lakhs at an interest rate of 8% per annum, compounded half-yearly. He repays Rs 10320 at the end of the first year and closes the loan by paying the outstanding amount at the end of the third year. Then, the total interest, in rupees, paid over the three years is nearest to

- A 45311
- B 51311
- C 33130
- D 40991

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Anil borrows Rs 2 lakhs at an interest rate of 8% per annum, compounded half-yearly. He repays Rs 10320 at the end of the first year and closes the loan by paying the outstanding amount at the end of the third year. Then, the total interest, in rupees, paid over the three years is nearest to

A) 45311
B) 51311
C) 33130
D) 40991

$200000 \times 1.04 = 208000$

$100000 \times 1.0816 = 108160$

216000

$P \left(1 + \frac{R}{100}\right)^N$

$P \left(1 + \frac{R/2}{100}\right)^{2N}$



VIDEO SOLUTION

Question 2

Anil invests Rs. 22000 for 6 years in a certain scheme with 4% interest per annum, compounded half-yearly. Sunil invests in the same scheme for 5 years, and then reinvests the entire amount received at the end of 5 years for one year at 10% simple interest. If the amounts received by both at the end of 6 years are same, then the initial investment made by Sunil, in rupees, is

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Answer: _____

$22000 \times 1.12 = 24640$

$P \left(1 + \frac{R}{100}\right)^N$

$P \left(1 + \frac{R/2}{100}\right)^{2N}$



VIDEO SOLUTION

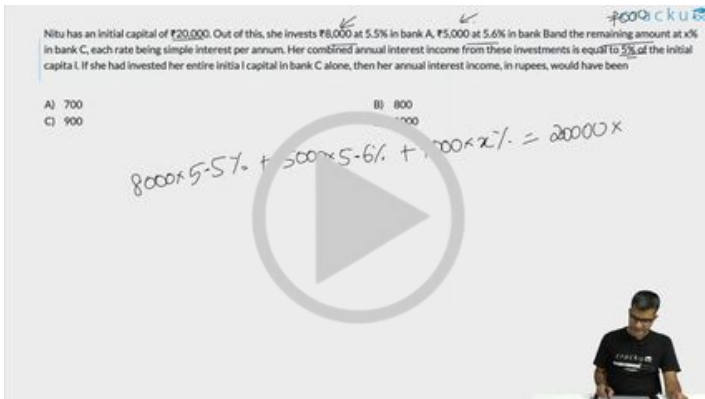
Question 3

Nitu has an initial capital of ₹20,000. Out of this, she invests ₹8,000 at 5.5% in bank A, ₹5,000 at 5.6% in bank B and the remaining amount at $x\%$ in bank C, each rate being simple interest per annum. Her combined annual interest income from these investments is equal to 5% of the initial capital. If she had invested her entire initial capital in bank C alone, then her annual interest income, in rupees, would have been

- A 700
- B 800
- C 900
- D 1000

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A) 700 B) 800
C) 900 D) 1000

$$8000 \times 5.5\% + 5000 \times 5.6\% + 7000 \times x\% = 20000 \times 5\%$$


VIDEO SOLUTION



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Question 4

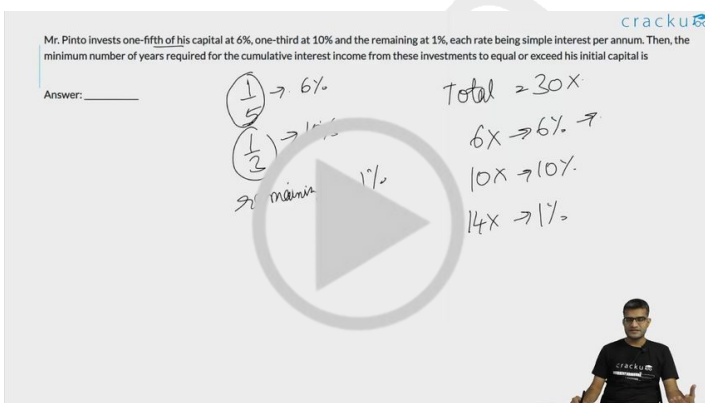
Mr. Pinto invests one-fifth of his capital at 6%, one-third at 10% and the remaining at 1%, each rate being simple interest per annum. Then, the minimum number of years required for the cumulative interest income from these investments to equal or exceed his initial capital is

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Answer: _____

$\frac{1}{5} \rightarrow 6\%$
 $\frac{1}{3} \rightarrow 10\%$
 2/5 remaining 1%

Total = $30x$
 $6x \rightarrow 6\%$
 $10x \rightarrow 10\%$
 $14x \rightarrow 1\%$



VIDEO SOLUTION

Question 5

Alex invested his savings in two parts. The simple interest earned on the first part at 15% per annum for 4 years is the same as the simple interest earned on the second part at 12% per annum for 3 years. Then, the percentage of his savings invested in the first part is

- A 37.5%
- B 62.5%
- C 60%
- D 40%

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A) 37.5%
B) 62.5%
C) 60%
D) 40%

Handwritten solution:

$$\frac{A \times 15 \times 4}{100} = \frac{B \times 12 \times 3}{100}$$

Video solution showing a man writing the solution.

VIDEO SOLUTION

Question 6

Bank A offers 6% interest rate per annum compounded half-yearly. Bank B and Bank C offer simple interest but the annual interest rate offered by Bank C is twice that of Bank B. Raju invests a certain amount in Bank B for a certain period and Rupa invests ₹ 10,000 in Bank C for twice that period. The interest that would accrue to Raju during that period is equal to the interest that would have accrued had he invested the same amount in Bank A for one year. The interest accrued, in INR, to Rupa is

- A 3436
- B 2436
- C 2346
- D 1436

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A) 3436
B) 2436
C) 2346
D) 1436

Handwritten solution:

Let Raju's investment be ₹ x for t years in Bank B.
 Interest in Bank B: $\frac{x \times r \times t}{100}$
 Interest in Bank A for 1 year: $x \left(1 + \frac{6}{200}\right)^2 - x$
 Equating the two interests:
 $\frac{x \times r \times t}{100} = x \left(1 + \frac{6}{200}\right)^2 - x$
 $\frac{r \times t}{100} = \left(1 + \frac{6}{200}\right)^2 - 1$
 $\frac{r \times t}{100} = \frac{101}{10000}$
 $r \times t = 101$
 Rupa's investment in Bank C: ₹ 10,000 for $2t$ years.
 Interest in Bank C: $\frac{10000 \times 2r \times 2t}{100} = 400rt$
 Substituting $rt = 101$:
 Interest = $400 \times 101 = 40400$ (This is incorrect, the handwritten solution shows a different path).

Video solution showing a man writing the solution.

VIDEO SOLUTION



Question 7

Anil invests some money at a fixed rate of interest, compounded annually. If the interests accrued during the second and third year are ₹ 806.25 and ₹ 866.72, respectively, the interest accrued, in INR, during the fourth year is nearest to

- A 929.48
- B 934.65
- C 931.72
- D 926.84

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A) 929.48
B) 934.65
C) 931.72
D) 926.84

Handwritten solution:

$$P(1+r)^2 - P(1+r) = 806.25$$

$$P(1+r)^3 - P(1+r)^2 = 866.72$$

$$P(1+r)^4 - P(1+r)^3 = ?$$

Let $P(1+r)^2 = X$
 $X - P(1+r) = 806.25$
 $X(1+r) - X = 866.72$
 $Xr = 866.72$
 $X = \frac{866.72}{r}$
 $\frac{866.72}{r} - P(1+r) = 806.25$
 $866.72 - P(1+r)r = 806.25r$
 $866.72 - P(1+r)r = 806.25r$
 $866.72 - P(1+r)r = 806.25r$

Video player interface with a play button and a person watching the video.

VIDEO SOLUTION

Question 8

For the same principal amount, the compound interest for two years at 5% per annum exceeds the simple interest for three years at 3% per annum by Rs 1125. Then the principal amount in rupees is

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Handwritten solution:

$$P \left(1 + \frac{5}{100}\right)^2 - P = P \times \frac{3}{100} \times 3 - P$$

$$P \left(1 + \frac{5}{100}\right)^2 - P = P \times \frac{9}{100} - P$$

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$$P \left(1 + \frac{5}{100}\right)^2 - P = P \times \frac{9}{100} - P$$

Video player interface with a play button and a person watching the video.

VIDEO SOLUTION

Question 9

A person invested a certain amount of money at 10% annual interest, compounded half-yearly. After one and a half years, the interest and principal together became Rs.18522. The amount, in rupees, that the person had invested is

cracku

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$I = 10\% \text{ p.a.} = 5\%$
 $T = 1.5 \text{ yr.} = 3 \text{ yr.}$

R

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Question 10

Veeru invested Rs 10000 at 5% simple annual interest, and exactly after two years, Joy invested Rs 8000 at 10% simple annual interest. How many years after Veeru's investment, will their balances, i.e., principal plus accumulated interest, be equal?

Veeru invested Rs 10000 at 5% simple annual interest, and exactly after two years, Joy invested Rs 8000 at 10% simple annual interest. How many years after Veeru's investment, will their balances, i.e., principal plus accumulated interest, be equal?

$T \rightarrow 10000$ 5%
 $T-2 \rightarrow 8000$ 10%
 $\frac{5}{4} 10000 \left(1 + \frac{5 \times T}{100} \right) = 8000 \left(1 + \frac{10 \times (T-2)}{100} \right)$

VIDEO SOLUTION

Question 11

A person invested a total amount of Rs 15 lakh. A part of it was invested in a fixed deposit earning 6% annual interest, and the remaining amount was invested in two other deposits in the ratio 2 : 1, earning annual interest at the rates of 4% and 3%, respectively. If the total annual interest income is Rs 76000 then the amount (in Rs lakh) invested in the fixed deposit was _____

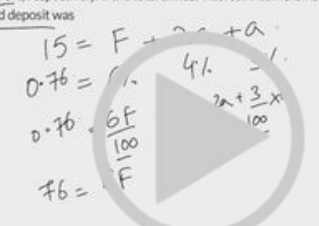
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$$15 = F + 2x + x$$

$$0.76 = \frac{6F}{100} + \frac{2x \cdot 4}{100} + \frac{x \cdot 3}{100}$$

$$76 = 6F + 8x + 3x$$

$$76 = 6F + 11x$$



VIDEO SOLUTION

Question 12

Amal invests Rs 12000 at 8% interest, compounded annually, and Rs 10000 at 6% interest, compounded semi-annually, both investments being for one year. Bimal invests his money at 7.5% simple interest for one year. If Amal and Bimal get the same amount of interest, then the amount, in Rupees, invested by Bimal is

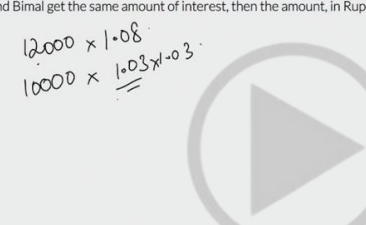
cracku

Amal invests Rs 12000 at 8% interest, compounded annually, and Rs 10000 at 6% interest, compounded semi-annually, both investments being for one year. Bimal invests his money at 7.5% simple interest for one year. If Amal and Bimal get the same amount of interest, then the amount, in Rupees, invested by Bimal is

$$12000 \times 1.08$$

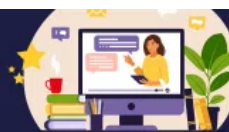
$$10000 \times 1.03 \times 1.03$$

$$P \times \left(1 + \frac{7.5}{100}\right)^n$$

$$\frac{n}{2} \times \left(1 + \frac{7.5}{100}\right)^{\frac{n}{2}}$$



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Question 13

Gopal borrows Rs. X from Ankit at 8% annual interest. He then adds Rs. Y of his own money and lends Rs. X+Y to Ishan at 10% annual interest. At the end of the year, after returning Ankit's dues, the net interest retained by Gopal is the same as that accrued to Ankit. On the other hand, had Gopal lent Rs. X+2Y to Ishan at 10%, then the net interest retained by him would have increased by Rs. 150. If all interests are compounded annually, then find the value of X + Y.

 VIDEO SOLUTION

John borrowed Rs. 2,10,000 from a bank at an interest rate of 10% per annum, compounded annually. The loan was repaid in two equal instalments, the first after one year and the second after another year. The first instalment was interest of one year plus part of the principal amount, while the second was the rest of the principal amount plus due interest thereon. Then each instalment, in Rs., is

 VIDEO SOLUTION

Answers

1. B	2. 20808	3. B	4. 20	5. A	6. B	7. C	8. 90000
9. 16000	10. 12	11. 9	12. 20920	13. 4000	14. 121000		

Explanations

1. **B**

It is given that Anil borrows Rs 2 lakhs at an interest rate of 8% per annum, compounded half-yearly. It is also known that he repays Rs 10320 at the end of the first year and closes the loan by paying the outstanding amount at the end of the third year.

The total amount at the end of the first year is: $200000 \times \frac{104}{100} \times \frac{104}{100} = 216320$

He repays 10320 rupees at the end of the first year, which implies the amount that remains unpaid at the end of the first year is 206000 rupees.

This unpaid amount will accrue interest for another two years.

Hence, the final amount at the end of three years is $206000 \times \frac{104}{100} \times \frac{104}{100} \times \frac{104}{100} = 240990.86$

Hence, the accrued interest in these two years is $(240990.86 - 206000) = 34990.86$ rupees.

Hence, the total interest accrued over the three years = $(34990.86 + 16320) = 51311$ rupees.

The correct option is B

 VIDEO SOLUTION

2. **20808**

Anil invested 22000 for 6 years at 4% interest compounded half-yearly

=> Amount = $22000 (1.02)^6$

Let Sunil invest 'S' rupees for 5 years at 4% C.I. half-yearly and 10% S.I. for 1 additional year

=> Amount = $S (1.02)^{10} (1.1)$

Given that the both amounts are equal

=> $22000 (1.02)^{12} = S (1.02)^{10} (1.1)$

=> $S = \frac{22000 (1.02)^2}{1.1} = 20808$

 VIDEO SOLUTION

3. **B**

It is given,

$$\frac{5.5 \times 1 \times 8000}{100} + \frac{5.6 \times 1 \times 5000}{100} + \frac{x \times 1 \times 7000}{100} = \frac{5}{100} \times 20000$$

$$440 + 280 + 70x = 1000$$

$$x = 4\%$$

$$\text{Interest} = \frac{20000 \times 4 \times 1}{100} = \text{Rs } 800$$

The answer is option B.

 VIDEO SOLUTION

4. **20**

Let the total investment be 15x and the no. of years required be T years

$$\frac{(3x \times 6 \times T)}{100} + \frac{(5x \times 10 \times T)}{100} + \frac{(7x \times 1 \times T)}{100} \geq 15x$$

$$\text{or, } \frac{75xT}{100} \geq 15x$$

$$\text{or, } T \geq 20$$

So minimum value of T is 20 years

 VIDEO SOLUTION

5. A

Let the savings invested in first part and second part be 'x' and 'y', respectively.

It is given,

$$\frac{x \times 15 \times 4}{100} = \frac{y \times 12 \times 3}{100}$$

$$60x = 36y$$

$$5x = 3y$$

$$\text{Required percentage} = \frac{x}{x+y} \times 100 = \frac{3}{3+5} \times 100 = 37.5\%$$

The answer is option A.

 VIDEO SOLUTION

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6. B

Bank A: 6% p.a. 1/2 yearly (CI)

Bank B: x% p.a (SI)

Bank C: 2x% p.a (SI)

Let Raju invest Rs P in bank B for t years. Hence, Rupa invests Rs 10,000 in bank C for 2t years.

Now,

$$P \left(\frac{x}{100} \right) t = P \left(1 + \frac{3}{100} \right)^2 - P$$

$$\left(\frac{x}{100} \right) t = 1.0609 - 1$$

$$\left(\frac{x}{100} \right) t = 0.0609$$

We need to calculate

$$SI = 10000 \times 2t \times \left(\frac{2x}{100} \right) = 40000 \left(\frac{x}{100} \right) t = 40000 \times 0.0609 = 2436$$

 VIDEO SOLUTION

7. C

Let the principal amount be P and the interest rate be r.

$$\text{Then } P(1+r)^2 - P(1+r) = 806.25 \text{ -(1)}$$

$$P(1+r)^3 - P(1+r)^2 = 866.72 \text{ -(2)}$$

Dividing (2) by (1), we get:

$$\frac{(P(1+r)^3 - P(1+r)^2)}{P(1+r)^2 - P(1+r)} = \frac{866.72}{806.25}$$

$$\frac{((1+r)^2 - 1 - r)}{1+r-1} = 1.075$$

$$\frac{r^2 + r}{r} = 1.075$$

$r=0.075$ or 7.5%

$$\frac{(\text{Interest accrued in 4th yr})}{(\text{Interest accrued in 3rd yr})} = \frac{X}{866.72}$$

$$\frac{(P(1+r)^4 - P(1+r)^3)}{P(1+r)^3 - P(1+r)^2} = \frac{X}{866.72}$$

Dividing numerator and denominator by $P(1+r)^2$

$$\frac{r^2 + 2r + 1 - 1 - r}{1 + r - 1} = \frac{X}{866.72}$$

$$r + 1 = \frac{X}{866.72}$$

$$X = 1.075 \times 866.72 = 931.72$$

 VIDEO SOLUTION

8. 90000

For two years the compound interest is $\frac{PR(1)}{100} + \frac{PR(1)}{100} \left(1 + \frac{PR(1)}{100}\right)$

For three years the simple interest is $\frac{9PR}{100}$

Now $R(1) = 5\%$ and $R = 3\%$

$$\text{Hence } \frac{5P}{100} + \frac{5P}{100} (1.05) - \frac{9P}{100} = 1125$$

$$\frac{-4P}{100} + \frac{5.25P}{100} = 1125$$

$$\frac{1.25P}{100} = 1125$$

Solving we get $P = 90000$

 VIDEO SOLUTION

9. 16000

Given,

Rate of interest = 10%

Since it is compounded half-yearly, $R = 5\%$

$n = 3$

$$\text{We know, } A = P \left(1 + \frac{R}{100}\right)^n$$

$$18522 = P (1 + 0.05)^3$$

$$\Rightarrow P = 16000$$

 VIDEO SOLUTION

10. 12

Let their individual Amounts be equal after 't' years. Let their initial investments amount to A_V and A_J ;

$$A_V = 10,000 \left(1 + \frac{5t}{100}\right) \text{ and } A_J = 8,000 \left(1 + \frac{10(t-2)}{100}\right)$$

$$\text{Equating both: } 10,000 \left(1 + \frac{5t}{100}\right) = 8,000 \left(1 + \frac{10(t-2)}{100}\right)$$

On simplifying both sides, we get: $15t = 180$; $t = 12$

 VIDEO SOLUTION

11.9

Assuming the amount invested in the ratio 2:1 was $200x$ and $100x$, then the fixed deposit investment = $1500000 - 300x$

Hence, the interest = $200x \times 4/100 = 8x$ and $100x \times 3/100 = 3x$

Interest from the fixed deposit = $(1500000 - 300x) \times 6/100 = 90000 - 18x$

Hence the total interest = $90000 - 18x + 8x + 3x = 90000 - 7x = 76000$

$\Rightarrow 7x = 14000 \Rightarrow x = 2000$

Hence, the fixed deposit investment = $1500000 - 300 \times 2000 = 900000 = 9 \text{ lakhs}$

 VIDEO SOLUTION

12. 20920

The amount with Amal at the end of 1 year = $12000 \times 1.08 + 10000 \times 1.03 \times 1.03 = 23569$

Interest received by Amal = $23569 - 22000 = 1569$

Let the amount invested by Bimal = $100b$

Interest received by Bimal = $100b \times 7.5/100 = 7.5b$

It is given that the amount of interest received by both of them is the same

$7.5b = 1569$

$b = 209.2$

Amount invested by Bimal = $100b = 20920$

 VIDEO SOLUTION

13. 4000

Amount of interest paid by Ishan to Gopal if the borrowed amount is Rs. $(X+Y) = \frac{10}{100} \times (X+Y) = 0.1(X+Y)$

Gopal also borrowed Rs. X from Ankit at 8% per annum. Therefore, he has to return Ankit Rs. $0.08X$ as the interest amount on borrowed sum.

Hence, the interest retained by gopal = $0.1(X+Y) - 0.08X = 0.02X + 0.1Y \dots (1)$

It is given that the net interest retained by Gopal is the same as that accrued to Ankit.

Therefore, $0.08X = 0.02X + 0.1Y$

$\Rightarrow X = (5/3)Y \dots (2)$

Amount of interest paid by Ishan to Gopal if the borrowed amount is Rs. $(X+2Y) = \frac{10}{100} \times (X+2Y) = 0.1X + 0.2Y$

In this case the amount of interest retained by Gopal = $0.1X + 0.2Y - 0.08X = 0.02X + 0.2Y \dots (3)$

It is given that the interest retained by Gopal increased by Rs. 150 in the second case.

$\Rightarrow (0.02X + 0.2Y) - (0.02X + 0.1Y) = 150$

$\Rightarrow Y = \text{Rs. } 1500$

By substituting value of Y in equation (2), we can say that $X = \text{Rs. } 2500$

Therefore, $(X+Y) = \text{Rs. } 4000$.

 VIDEO SOLUTION

14. 121000

We have to equate the installments and the amount due either at the time of borrowing or at the time when the entire loan is repaid. Let us bring all values to the time frame in which all the dues get settled, i.e, by the end of 2 years.

John borrowed Rs. 2,10,000 from the bank at 10% per annum. This loan will amount to $2,10,000 \times 1.1 \times 1.1 = \text{Rs. } 2,54,100$ by the end of 2 years.

Let the amount paid as installment every year be Rs.x.

John would pay the first installment by the end of the first year. Therefore, we have to calculate the interest on this amount from the end of the first year to the end of the second year. The loan will get settled the moment the second installment is paid.

$$\Rightarrow 1.1x + x = 2,54,100$$

$$2.1x = 2,54,100$$

$$\Rightarrow x = \text{Rs. } 1,21,000.$$

Therefore, 121000 is the correct answer.

 VIDEO SOLUTION